The examination is being carried out on the **following application documents**

Description, Pages

1-44 as originally filed

Claims, Numbers

1-15 as originally filed

Drawings, Sheets

1/11-11/11 as originally filed

1 Unity of invention

1.1 The present application lacks unity (Article 82 EPC), as the following separate inventions or groups of inventions are not so linked as to form a single general inventive concept:

1. Invention 1: claims 1-5, 8, 9, 14 and 15: directed to generating the first radar image by masking a range-angle image (RAI) or range-Doppler image (RDI) based on an output of an order statistics (OS) constant false alarm rate (CFAR) (OS-CFAR) detector;
2. Invention 2: claims 6 and 7: directed to generating a heatmap with a tracker of the radar processing chain, and wherein the first radar processing output comprises the heatmap;
3. Invention 3: claims 10-13: directed to after training the hyperparameter selection neural network using imitation learning, training the hyperparameter selection neural network using reinforcement learning.

1.2 In order to assess unity, the subject-matter of the claims is compared to D1 which is considered to represent the closest prior art for the first invention.

The common matter between the two groups of invention is at most the method defined by the independent claim 1.

D1 discloses all features of claim 1, as follows:

A method comprising:

A. receiving first radar data from a millimeter-wave radar sensor (fig. 9: "*906*"

and para. [0023], where the object recognition implies a millimeter-wave radar);

B. receiving a set of hyperparameters with a radar processing chain (fig. 9: "*904-906*");

C. generating a first radar processing output using the radar processing chain based on the first radar data and the set of hyperparameters (fig. 9: "*902*");

D. updating the set of hyperparameters based on the first radar processing output using a hyperparameter selection neural network (fig. 9: "*904-906*");

E. receiving second radar data from the millimeter-wave radar sensor (fig. 9: "*906*"); and

F. generating a second radar processing output using the radar processing chain based on the second radar data and the updated set of hyperparameters (fig. 9: "*902*").

1.2.1 Claims 14 and 15 correspond essentially to claim 1; thus, the same applies mutatis mutandis.

1.2.2 As the subject-matter of claim 1 is anticipated by D1, it is common practice to examine the additional features of the dependent claims in order to identify possible special technical features according to Rule 44(1) EPC, which enable a decision to be reached on the issue of whether a single general inventive concept is present in the application (Article 82 EPC).

D1 also discloses or renders obvious the features of claims 2, 3, 8 and 9, which do therefore not define a contribution over the prior art:

– claims 2 and 3: D1, para. [0077]; the number of frames being lower or equal to 32 is a normal design option,

– claims 8 and 9: D1 (para. [0082]); the use of imitation learning is a straightforward alternative.

1.3 Special technical features

1. The first features that make a contribution over D1 are those of claim 4. They are "special technical features" STF1 within the meaning of R. 44(1) EPC. The technical effect of generating the first radar image by masking a range-angle image or range-Doppler image based on an output of an order statistics constant false alarm rate detector is to improve performance versus noise. The corresponding technical problem is to improve perfor-

mance versus noise.

2. The features of claim 6 are also not anticipated by D1. They are "special technical features" STF2 within the meaning of R. 44(1) EPC. The technical effect of generating a heatmap with a tracker of the radar processing chain, and wherein the first radar processing output comprises the heatmap is to enable elaboration of tracking data by the neural network. The corresponding technical problem is how to enable elaboration of tracking data by the neural network;

3. The features of claim 10 are also not anticipated by D1. They are "special technical features" STF3 within the meaning of R. 44(1) EPC. The technical effect of after training the hyperparameter selection neural network using imitation learning, training the hyperparameter selection neural network using reinforcement learning is to improve learning efficiency. The corresponding technical problem is to improve learning efficiency.

1.4 When comparing the special technical features STF1 - STF3, it is clear that they are not the same. Moreover, the corresponding technical problems of the different inventions:

1. improve performance versus noise,
2. enable elaboration of tracking data by the neural network,
3. improve learning efficiency,

are different. The special technical features STF1 - STF3 are thus not corresponding. The subject matter of the different inventions is therefore not linked by a common special technical feature. As a consequence, the application lacks unity (Article 82 and Rule 44 EPC).

2 The applicant replied to the invitation to pay additional search fees with letter dated and received on 06.02.2023, where it is submitted that the European search report is also to cover one further invention of claims 6 and 7.

3 Documents

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 US 2018/348343 A1 (ACHOUR MAHA [US] ET AL) 6 December 2018

D2 CHEIKH K ET AL: "Application of neural networks to radar signal detection in E1K /E1-distributed clutter", IEE PROCEEDINGS: RADAR, SONAR & NAVIGATION, INSTITUTION OF ELECTRICAL ENGINEERS, GB, vol. 153, no. 5, 19 October 2006, pages 460-466, XP006027295, ISSN: 1350-2395, DOI: 10.1049/IP-RSN:20050103

D3 US 2020/272148 A1 (KARASEV VASILIY [US] ET AL) 27 August 2020

D4 US 2021/116540 A1 (SANTRA AVIK [DE] ET AL) 22 April 2021

4 Clarity

The application does not meet the requirements of Article 84 EPC.

4.1 The term *"hyperparameter"* used in claim 1 is vague and unclear and leaves the reader in doubt as to the meaning of the technical feature to which it refers, thereby rendering the definition of the subject-matter of said claim unclear (Article 84 EPC): generally, the term hyperparameter is used in the contest of neural networks to indicate a parameter whose value is used to control the learning process (e.g. learning rate). Here, however, it seems that the term is used to refer to parameters for radar detection (see also claim 5). The claim should specify the meaning of this term.

4.2 The same applies mutatis mutandis to claim 14.

5 Novelty and Inventive Step

Furthermore, notwithstanding the above-mentioned lack of clarity, the present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claims 1, 2, 8, 14, 15 is not new within the meaning of Article 54(1) and (2) EPC and the subject-matter of claims 3-5 and 9 does not involve an inventive step within the meaning of Article 56 EPC.

5.1 Claim 1

5.1.1 Document D1 discloses:

A method comprising:

A. receiving first radar data from a millimeter-wave radar sensor (fig. 9: "*906*" and para. [0023], where the object recognition implies a millimeter-wave radar);

B. receiving a set of hyperparameters with a radar processing chain (fig. 9: "*904-906*");

C. generating a first radar processing output using the radar processing chain based on the first radar data and the set of hyperparameters (fig. 9: "*902*");

D. updating the set of hyperparameters based on the first radar processing output using a hyperparameter selection neural network (fig. 9: "*904-906*");

E. receiving second radar data from the millimeter-wave radar sensor (fig. 9: "*906*"); and

F. generating a second radar processing output using the radar processing chain based on the second radar data and the updated set of hyperparameters (fig. 9: "*902*").

The subject-matter of claim 1 is therefore not new (Article 54(1) and (2) EPC).

5.2 Claims 2, 3

D1, para. [0077]. The number of frames being lower or equal to 32 is a normal design option.

5.3 Claims 4 and 5

The features of claim 4 have the technical effect and solve the problem of improving performance versus noise. Document D2 shows the advantages of using an ANN-OS-CFAR in an environment with interfering targets (cf. page 464, second column "*To allow the neuronal detector with MLP(BP)...*" to page 465 "*...compared to the OS-CFAR detector*"). It would therefore be obvious to the person skilled in the art to apply these features with corresponding effect to the method of D1, thus arriving at a method according to claim 4. For data consistency, it follows that also the hyperparameters comprise the probability of false alarm hyperparameter, the number of reference cells hyperparameter, and the number of guard cell hyperparameter (claim 5).

5.4 Claims 8 and 9

Training the hyperparameter selection neural network, in particular using imitation learning: D1 (para. [0082]) shows reinforcement learning. The use of imitation learning (claim 9) is a straightforward alternative.

5.5 Claims 14 and 15

The subject matter of claims 14 and 15 differs in essence from that of the corresponding claim 1 by the type of claim, apparatus rather than method claim; therefore the objections raised for claim 1 apply mutatis mutandis equally for claims 14 and 15.

5.6 Claim 6

The method of any one of claims 1 to 5, further comprising, generating a heatmap with a tracker of the radar processing chain, and wherein the first radar processing output comprises the heatmap.

5.6.1 The technical effect and problem to be solved by the features of claim 6 can be seen as: to enable elaboration of tracking data by the neural network.

5.6.2 Document D1 does not disclose these features and only shows a multi-object tracker (cf. fig. 9 "*912*"), which is, however, not used to update the set of hyperparameters. Starting from D1, the skilled person would find no hint to use the output of said tracker for "*updating the set of hyperparameters based on the first radar processing output using a hyperparameter selection neural network*" (i.e. to use its output as input for the neural network). Known solutions involve a separate neural network trained and working with tracking data, which would first be considered by the skilled person. An example is provided by D3, para. [0051], that relates to an automotive sensor system using neural network(s) in particular for determining and predicting the trajectory of an object (cf. para. [0008]). Taking into account D3 the skilled person would, thus, not be prompted with the proposed solution. Para. [0016] mentions that "*the machine learning model can output a heat map associated with prediction probabilities ... the detected pose of the pedestrian (appearance and/or geometric) and/or the output velocity of the pedestrian may provide additional features for input into such additional machine learned models, the result of which comprising additional data for determining likely positions of the pedestrian in time (such as may be represented in a heat map as output by one or more of the machine learned models). In some examples, at least one predicted trajectory can be determined based at least in part on the heat map. Additionally, output of a machine learning (e.g., trajectory prediction) model based on image data can be used as input into a heatmap generation model*". Thus, D3 suggests the idea of a heatmap as output of a neural network and, additionally, of a heatmap generation model that receives trajectory predictions from a neural network, but it does not refer to "*generating a heatmap with a tracker of the radar processing chain, and where-*

in the first radar processing output comprises the heatmap [and] *updating the set of hyperparameters based on the first radar processing output using a hyperparameter selection neural network*".

The subject-matter of the claim is therefore considered to involve an inventive activity (Art. 56 EPC).

5.7 Claim 7

The method of any one of claims 1 to 6, wherein the radar processing chain comprises a tracker, wherein the set of hyperparameters comprises a gating threshold hyperparameter, and wherein receiving the set of hyperparameters comprises receiving the gating threshold hyperparameter with the tracker, wherein optionally the tracker comprises an unscented Kalman filter: the same considerations as to claim 6 put forward in sections 5.6.1 and 5.6.2 above apply also to claim 7, since the introduction of the tracker in the radar processing chain as defined in claim 1 could not be considered straightforward in D1.

The subject-matter of the claim is therefore considered to involve an inventive activity (Art. 56 EPC).

6 If the applicant decides to proceed further with this application the following points should also receive attention:

I. the independent claims should be cast in correct two-part form (Rule 43(1) EPC).

II. reference signs in parentheses should be inserted in the claims (Rule 43(7) EPC). This applies to both the preamble and the characterising portion.

III. in the claims, only the reference signs should be inserted in parentheses (Rule 43(7) EPC).

IV. documents cited in section 1 above should be referred to in appropriate terms in the description (Rule 42(1)(b) EPC).

V. the description should be brought into line with the new claims (Rule 42(1)(c) EPC).

VI. any inconsistency between the claims and the description that leads to doubt concerning the matter for which protection is sought, thereby rendering the claims unclear (Article 84 EPC) should be removed, e.g. by indicating in the description that the embodiment/s concerned do/does not

form part of the invention (see Guidelines F-IV, 4.3(iii)). Failure to do so could lead to the refusal of the application under Article 97(2) EPC in combination with Article 84 EPC.

VII. the technical field of the invention (Rule 42(1)(a), GL F-II.4.2) must correspond to the amended set of claims.

VIII. the summary of the invention (Rule 42(1)(c) and GL F-II.4.5) must correspond to the amended set of claims. If appropriate, statements such as "The invention is set out in the appended set of claims" or "The invention is as defined in claim X" can be applied instead of repeating the claims verbatim.

IX. all technical features of the independent claims shall not be presented as optional in the description. Terms such as "for example", "may", "can", "exemplary", "optionally", "preferably" and the like should be removed when preceding a feature of an independent claim.

X. in amending the application the applicant should take care to avoid giving rise to objection under Article 123(2) EPC by the inadvertent addition or deletion of subject matter which extends the content of the application beyond that of the application as filed.

XI. in order to facilitate the examination of the conformity of the amended application with the requirements of Article 123(2) EPC, the applicant is requested to clearly identify the amendments carried out, irrespective of whether they concern amendments by addition, replacement or deletion, and to indicate the passages of the application as filed on which these amendments are based (cf. Rule 137(4) EPC, Guidelines H-III, 2.1).

XII. any vague and imprecise statement in the description that implies that the subject-matter for which protection is sought may be different to that defined by the claims, thereby resulting in a lack of clarity of the claims (Article 84 EPC) when the description is used to interpret the claims (see Guidelines F-IV, 4.4), should be amended to remove this inconsistency.